

EXPERIMENT 6b

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize customer data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Algorithm 1

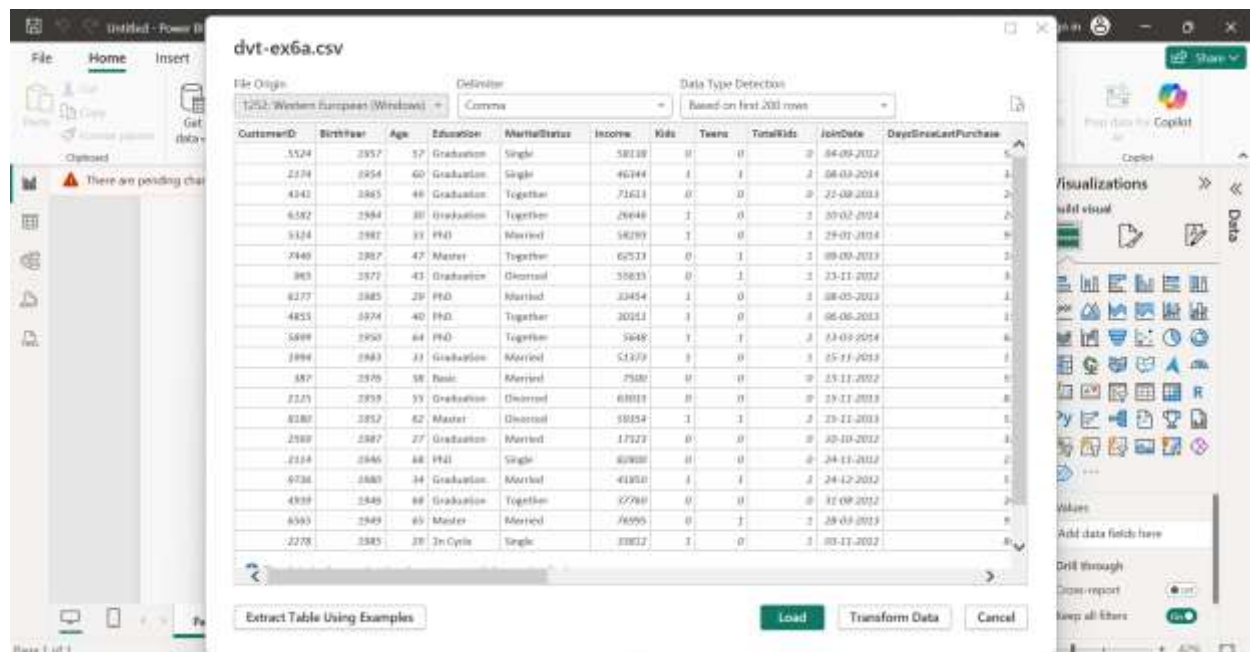
Importing Data into Power BI

Step 1 Open Power BI Desktop

Step 2 Select: Home → Get Data

Step 3 Choose CSV

Step 4 Browse dataset and select it



Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove blank rows

Step 4 Standardize the Education column

Step 5 Handle null values

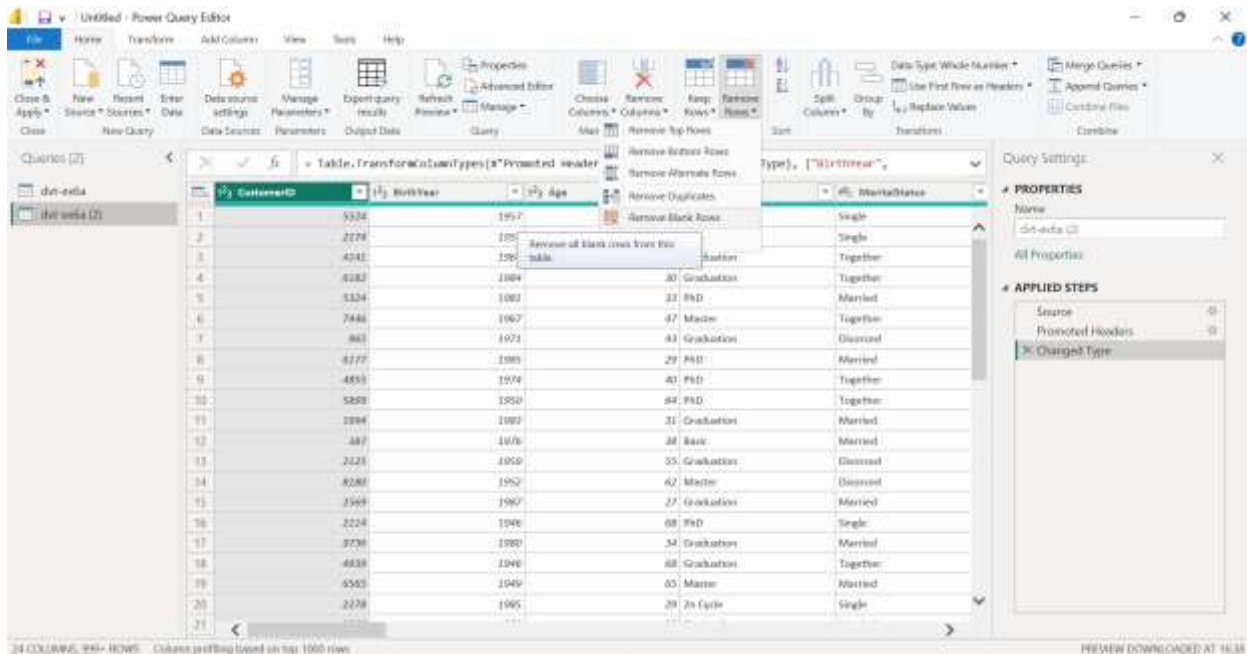
Step 6 Apply changes

Procedure

1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.

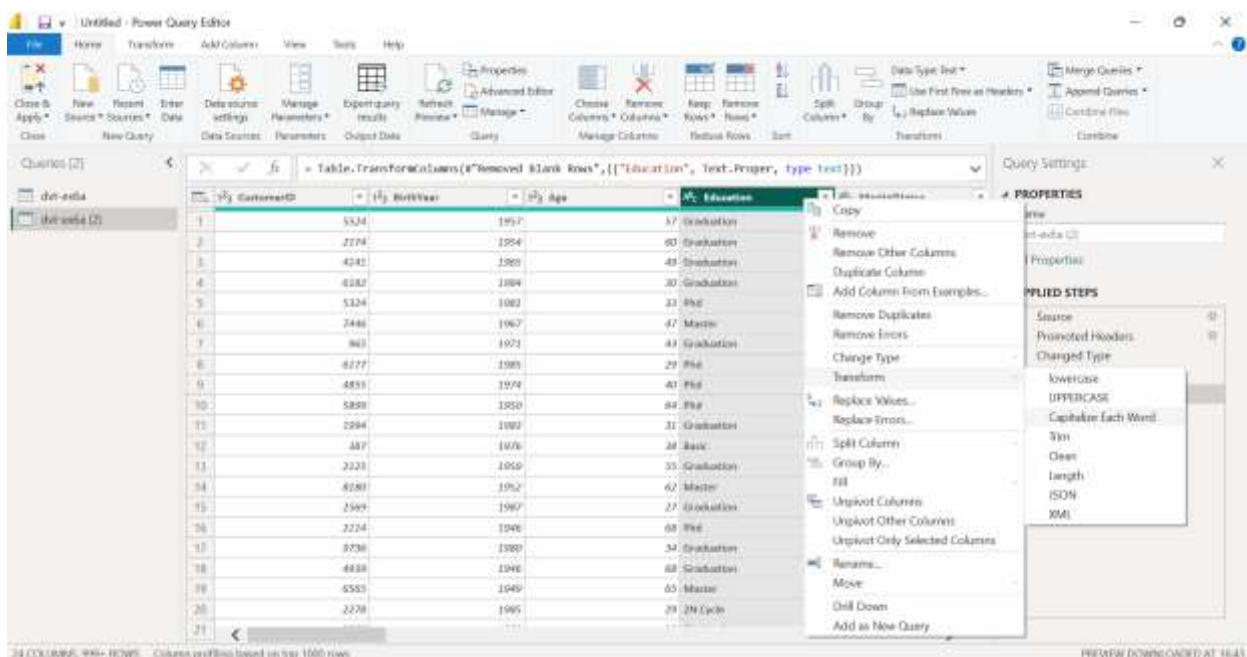
2. In the Power Query Editor, examine the imported table and verify that all required columns are present, such as CustomerID, BirthYear, Age, Education, MaritalStatus, Income, Kids, Teens, TotalKids, JoinDate, and DaysSinceLastPurchase.

3. Remove blank rows from the dataset by selecting Home → Remove Rows → Remove Blank Rows. This removes completely empty records that could interfere with analysis and calculations.



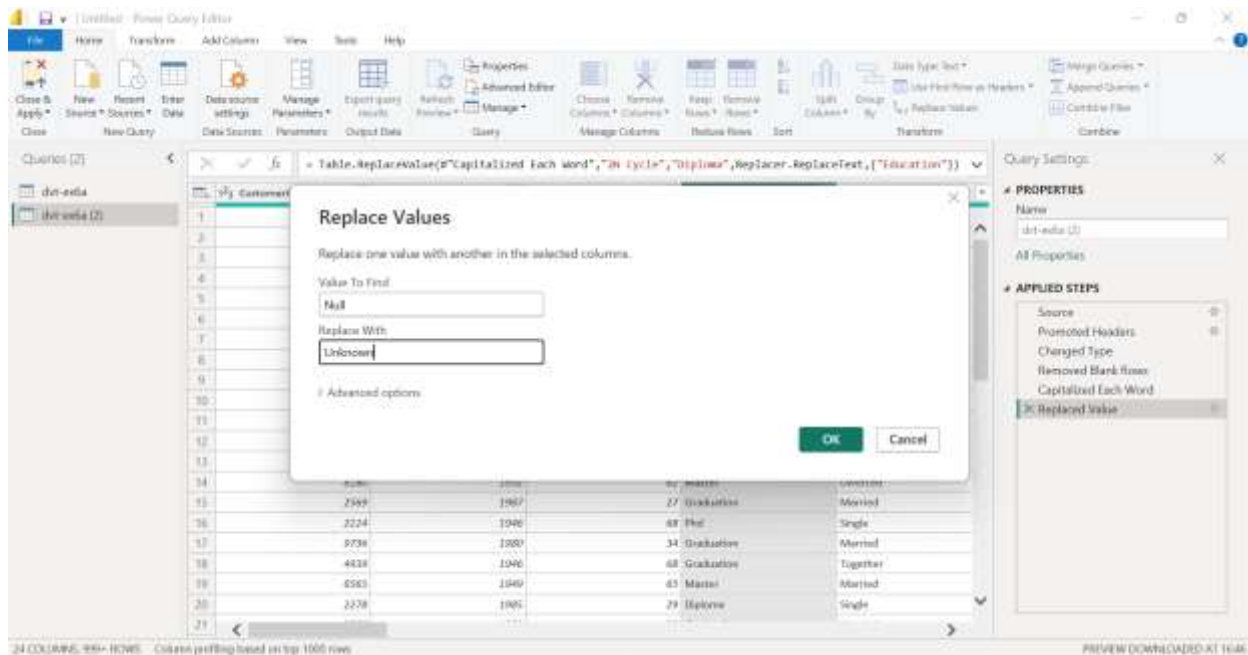
4. Remove duplicate records by selecting all columns in the table using Ctrl+A on the column headers and choosing Home → Remove Rows → Remove Duplicates. Power Query will remove records that contain identical values across all selected columns.

5. Standardize the values in the Education column. Right-click the Education column, select Transform, and choose Capitalize Each Word. This ensures that category labels are represented consistently.



6. Correct an inconsistent category label in the Education column. Using Replace Values, the entry “2N Cycle” is replaced with “Diploma” so that the education categories used across the report are consistent and easy to interpret.

7. Handle missing values in the Education column. Right-click Education, select Replace Values, and replace blank or null values with “Unknown”.



8. Verify and correct the data types of all columns. Set CustomerID, BirthYear, Age, Kids, Teens, and TotalKids to Whole Number; Income to Decimal Number; JoinDate to Date; and Education and MaritalStatus to Text.

9. Review the Applied Steps pane on the right side of the Power Query Editor. Confirm that all transformation steps (Removed Blank Rows, Capitalized Each Word, Replaced Value, and the null-handling step) have been recorded in the correct order.

10. After completing all cleaning and transformation operations, select Home → Close & Apply. Power BI will apply the transformations and load the cleaned data into the data model.

Algorithm 3

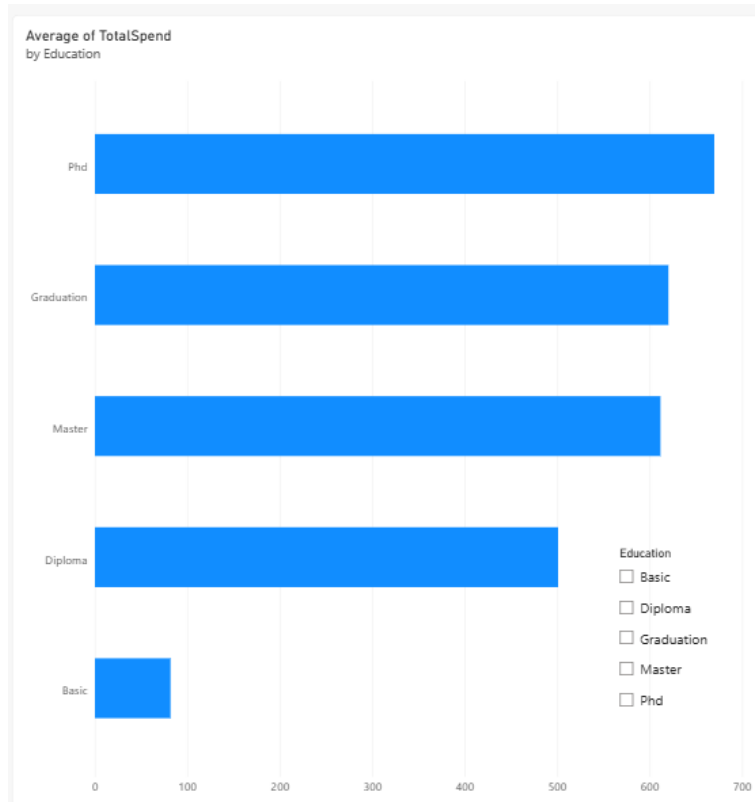
Creating a Bar Chart

Step 1 Choose Bar Chart Visualization

Step 2 Drag Education → Y-Axis

Step 3 Drag TotalSpend → X-Axis (set to Average)

Step 4 Format chart



Interpretation:

Customers with a PhD have the highest average spend, followed closely by Graduation and Master's degree holders.

Customers with only Basic education have by far the lowest average spend, well below all other education levels.

Algorithm 4

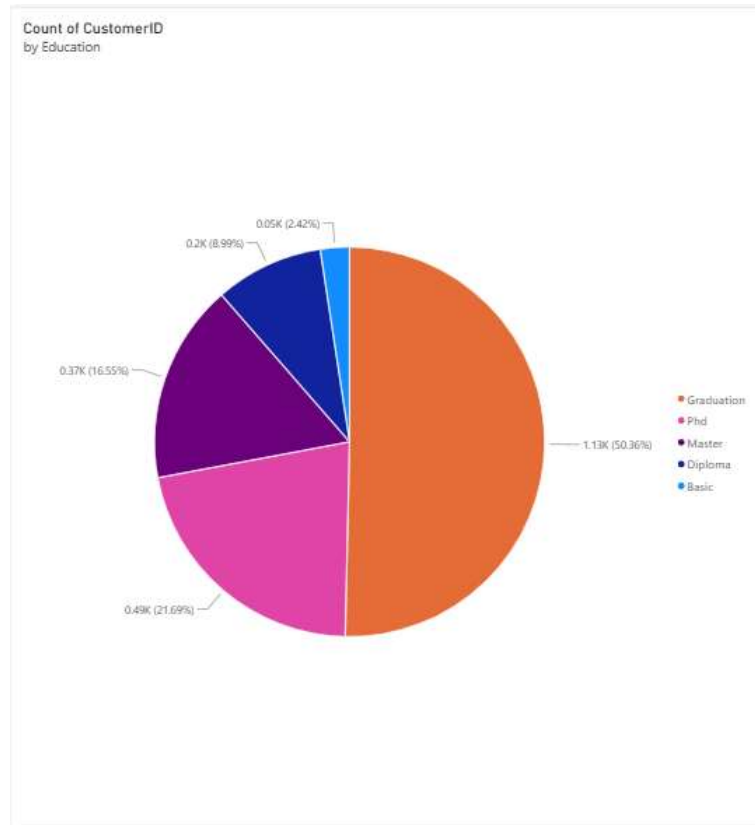
Creating a Pie Chart

Step 1 Select Pie Chart

Step 2 Drag Education → Legend

Step 3 Drag CustomerID → Values (set to Count)

Output:



Interpretation:

Graduation is the largest customer segment, making up just over 50% (1.13K) of all customers.

PhD holders form the next largest group at about 21.7% (0.49K), followed by Master's at 16.6% (0.37K) and Diploma at 9.0% (0.2K).

Basic education is the smallest segment at roughly 2.4% (0.05K) of customers.

Algorithm 5

Creating the Dashboard

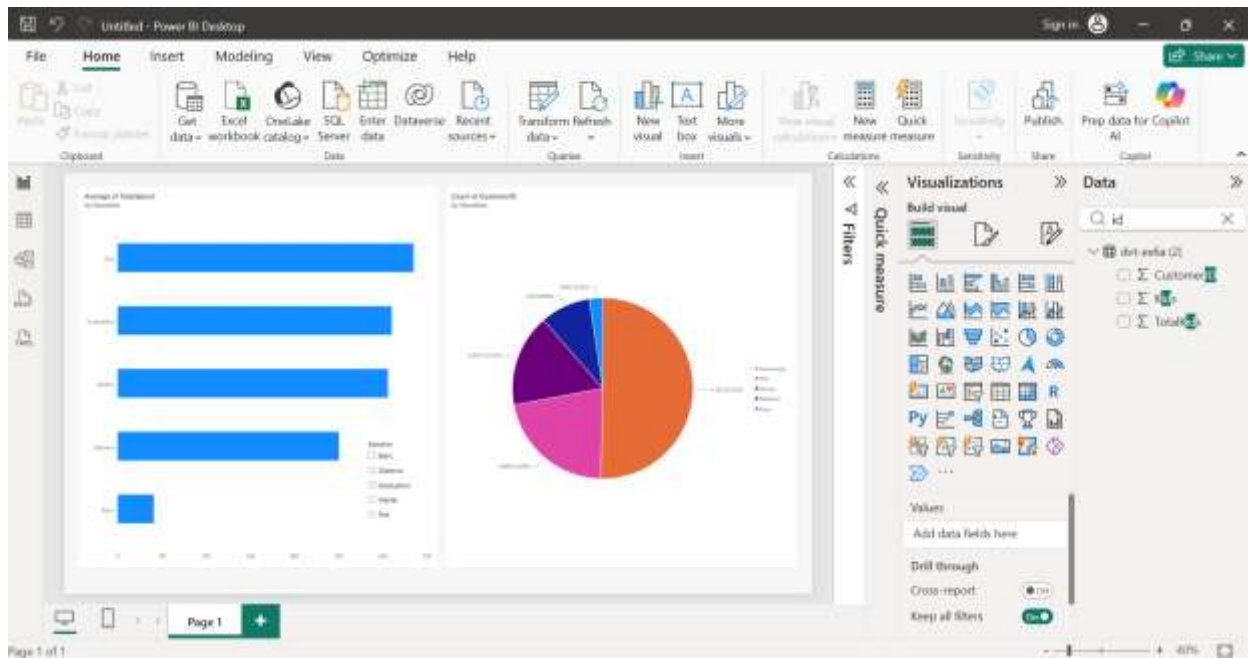
Step 1 Add Bar Chart (Average of TotalSpend by Education)

Step 2 Add Pie Chart (Count of CustomerID by Education)

Step 3 Arrange Visuals on the report page

Step 4 Save Dashboard

Output:



Result

The customer data was successfully imported, transformed, analyzed, and visualized in Power BI Desktop, and an interactive dashboard combining a bar chart and a pie chart was created for effective decision-making.

231401014